Don't just read about data structures. Build them, test them and time them. Every structure in this book is a working class you write yourself, checked against Python's built-ins on thousands of random inputs and measured with real benchmarks, so you see Big O happen on your own screen.

Build arrays, linked lists, hash tables, heaps, search trees, an AVL tree, a trie, union-find and graphs; implement eight sorts, Dijkstra, Kruskal and more. Then master the interview patterns: two pointers, sliding window, prefix sums, BFS and DFS, dynamic programming, greedy and backtracking.

Every listing was run on Python 3.12 to 3.15. Every chapter has real-world uses and exercises with tested solutions.

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DATA STRUCTURES AND ALGORITHMS IN PYTHON, HANDS-ON

AADHAVAN MUTHURENGAN

tasks.py

```python
import heapq

q = [(3, "ship"), (1, "code"), (2, "test")]
heapq.heapify(q)
while q:
    print(heapq.heappop(q))
```

```
$ python tasks.py
(1, 'code')
(2, 'test')
(3, 'ship')
```

- ARRAYS
- HASH MAPS
- TREES
- GRAPHS
- PATTERNS
- DP

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BUILT, TESTED AND TIMED

Build Every Structure Yourself, Benchmark It, and Master the Interview Patterns

BUILT FROM SCRATCH • REAL BENCHMARKS • INTERVIEW PATTERNS

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